MIP-BENCH: A Benchmark for Instructional Grounding in Medical Imaging Presentations

Md Zabirul Islam[1], **Md Motaleb Hossen Manik**[1], **Jianxu (Jayson) Wang**[2], **Yuxuan Liang**[2], **Ge Wang**[2,†]
[1]Department of Computer Science, Rensselaer Polytechnic Institute
[2]Department of Biomedical Engineering, Rensselaer Polytechnic Institute
{islamm11, manikm, wangj68, liangy15, wangg6}@rpi.edu
[†]Corresponding author

Abstract

We introduce a *reusable pipeline* for evaluating whether vision–language models (VLMs) recover the *instructional intent* of a lecture slide from its visual content alone—a property distinct from semantic or visual correctness and critical for educational deployment. The pipeline converts aligned instructor narration into a structured concept graph $\mathcal{I}(\mathcal{S}) = \langle \mathcal{C}, \mathcal{R}, \mathcal{B} \rangle$ of target concepts, directed relations, and scope boundary, and scores VLM responses with the *Instructional Grounding Score* (IGS), a weighted composite of Concept Recall (CR), Relational Validity (RV), and Scope Fidelity (SF). The narration-alignment module, the concept-graph formalization, the extraction prompts, and the ground-truth-conditioned judge are all domain-independent; only the terminology list is curriculum-specific. To instantiate and stress-test the pipeline, we build MIP-BENCH, a benchmark of 1,117 slides from a complete medical imaging curriculum spanning mathematical foundations, signal processing, physics, and computational methods across 23 lectures. Extraction and judging use GPT-4o—a model outside the evaluated family—and IGS is cross-validated against a structurally independent graph-based metric ($r = 0.41$–0.54), a second non-OpenAI LLM judge (Anthropic `claude-opus-4-6`) with per-model Pearson $r_{\text{IGS}} = 0.694$–0.835 and *identical* cross-provider model ordering (Kendall $\tau = 1.0$), and pooled human annotators ($r = 0.729$ on overall IGS, 88% of slides within 0.2). Across five open-weight VLMs from 7B to 40B parameters, the best model reaches IGS $= 0.642$, well below pooled human performance (IGS $= 0.690$); all five exhibit a consistent scope–concept gap (SF $-$ CR $= +0.10$ to $+0.14$), and two supervised LoRA fine-tuning strategies fail to close it. We further show that model rankings are stable when the extractor is simulated to lose up to 30% of its gold concepts (mean rank Spearman $\bar{\rho} = 0.94$, max flip = 1), defending the benchmark under imperfect extraction. We release the pipeline, extraction prompts, splits, and judge protocol so new structured scientific curricula (physics, chemistry, biology) can be instrumented with a $\sim$50-slide audit; MIP-BENCH is the first instantiation.

1 Introduction

Instructional grounding is not semantic correctness. Vision–language models have achieved strong results on medical visual question answering [1, 2], radiology report generation [3, 4, 5], and slide-based question answering [6]. Evaluation in these settings asks whether the model's output is factually consistent with the visible content. For educational deployment, factual consistency is insufficient.

Table 1: Evaluation criteria across representative VLM benchmarks. ✓ = explicitly evaluated; ○ = partially; × = not evaluated. MIP-BENCH is the only benchmark surveyed that evaluates instructional intent as a first-class target.

Criterion	VQA-Med	PathVQA	SlideVQA	CARES	MedSG	MIP-BENCH
Factual accuracy	✓	✓	✓	✓	✓	✓
Visual comprehension	✓	✓	○	○	✓	✓
Concept identification	×	×	○	×	×	✓
Relational validity	×	×	×	×	×	✓
Scope fidelity	×	×	×	×	×	✓
Instructional intent	×	×	×	×	×	✓

We study *instructional grounding*: whether a model's response reflects what a slide is *designed to teach*—its target concepts, the intended relations among those concepts, and the scope boundary that defines what the slide covers and what it does not. Consider a slide introducing the convolution theorem. A model that responds, "This slide shows mathematical notation and a signal processing diagram," is semantically accurate but instructionally hollow: it identifies no target concept. A model that correctly defines convolution but drifts into uncovered applications violates the scope boundary without making a factual error. Both failure modes are invisible to standard metrics, yet either can mislead a student using the model for study support.

Why existing benchmarks do not address this gap. Recent medical VLM benchmarks have made important progress on *clinical* evaluation dimensions: CARES [7] benchmarks trustworthiness across safety, fairness, and privacy; MedSG-Bench [8] evaluates spatial grounding of pathological regions across image sequences; and Micro-Bench [9] tests domain-specific visual understanding in microscopy. None addresses the distinct challenge of *educational grounding*: whether a model's response encodes the specific concepts a slide intends to teach, the relational structure among those concepts, and the scope boundary set by the instructor. Earlier benchmarks such as PathVQA [10], VQA-Med [11, 12], and SlideVQA [13] similarly evaluate factual retrieval or visual comprehension without modeling instructional intent. Table 1 summarizes these distinctions.

The benchmark: a controlled curriculum testbed. We introduce MIP-BENCH, constructed from a complete medical imaging curriculum comprising 1,117 slides across 23 lectures. The benchmark forms a controlled scientific reasoning testbed in which foundational concepts recur and compose across lectures: the curriculum spans mathematical foundations (Fourier analysis, linear systems), physics principles (X-ray attenuation, MRI relaxation), computational methods (filtered back-projection, neural network reconstruction), and clinical modalities (CT, MRI, PET, ultrasound). Cross-cutting concepts such as the Fourier transform (58 slides, 11 lectures) and convolution (46 slides, 8 lectures) support analysis of hierarchical grounding—whether a model that grounds a concept in isolation also grounds it correctly when it participates in a larger conceptual chain. Each slide is paired with a structured concept graph derived from aligned instructor narration, encoding target concepts, directed relational links, and an explicit scope boundary. Figure 1 illustrates the pipeline.

The metric: Instructional Grounding Score. We evaluate responses using IGS, a weighted composite of CR, RV, and SF with weights 0.40, 0.35, and 0.25. The weight ordering $w_{\text{CR}} > w_{\text{RV}} > w_{\text{SF}}$ encodes a pedagogical hierarchy: missing a target concept is the most direct grounding failure; incorrect relations distort concept connections; scope drift is least harmful. An ablation over 171 weight combinations confirms model rankings are robust to weight perturbation (mean Spearman $\rho = 0.998$), so the weight ordering rather than the exact magnitudes drives our conclusions. Section 2.2 gives the formal definition, and Section 4 describes implementation and validation.

Key findings. We evaluate five open-weight VLMs spanning 7B–40B parameters: Qwen2-VL-7B [14], InternVL2-8B/26B/40B [15], and LLaVA-1.6-34B [16].

1. **No model crosses IGS $= 0.642$.** The best model (Qwen2-VL-7B) is substantially below reliable grounding; MIP-BENCH discriminates clearly across models (ΔIGS $= 0.141$ between best and worst).

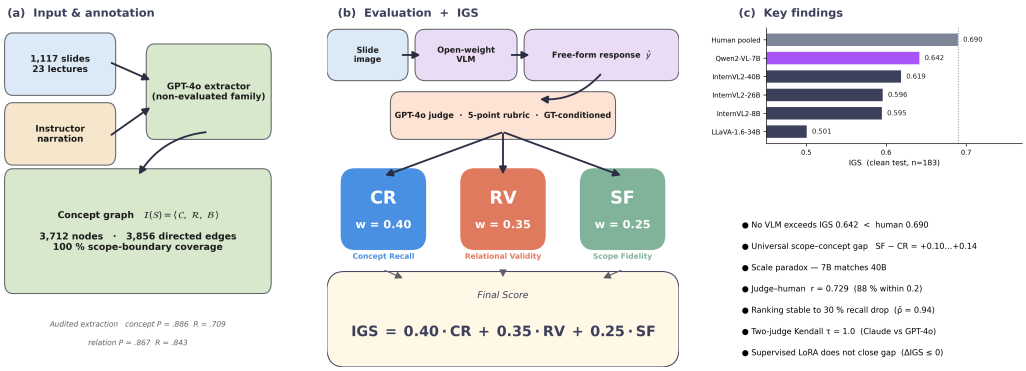

Figure 1: **MIP-BENCH pipeline.** **(a)** Each slide is paired with structured instructional intent $\mathcal{I}(S) = \langle \mathcal{C}, \mathcal{R}, \mathcal{B} \rangle$ extracted from aligned instructor narration via GPT-4o. **(b)** Open-weight VLMs produce free-form explanations scored by a GPT-4o judge against $\langle \mathcal{C}, \mathcal{R}, \mathcal{B} \rangle$ on three dimensions (CR, RV, SF), yielding $\text{IGS} = 0.40 \cdot \text{CR} + 0.35 \cdot \text{RV} + 0.25 \cdot \text{SF}$. **(c)** No model exceeds $\text{IGS} = 0.642$; all five exhibit a scope–concept gap ($\text{SF} - \text{CR} = +0.10$ to $+0.14$); pooled human–judge correlation is $r = 0.729$ ($n = 3$ annotators, 50 slides).

2. **Universal scope–concept gap.** SF exceeds CR by $+0.10$–$+0.14$ for all five models, indicating models perform implicit topic classification rather than explicit concept grounding—a qualitatively shallower operation invisible to standard similarity metrics.
3. **Scale does not uniformly help.** Qwen2-VL-7B matches or exceeds InternVL2-40B (0.642 vs. 0.619, CIs overlap) despite having 33B fewer parameters; centrality analysis identifies cross-cutting concepts as a bottleneck that added capacity does not resolve.
4. **Failure profiles overlap but differ across models.** Per-slide Pearson $r = 0.60$–0.82 across model pairs indicates models share common hard slides while also failing on distinct subsets.
5. **Supervised fine-tuning does not close the gap.** Teacher-narration LoRA ($\Delta\text{IGS} = -0.007$) and claims-structured LoRA ($\Delta\text{IGS} = -0.161$) both underperform the zero-shot baseline, suggesting simple supervised imitation is insufficient for instructional grounding.
6. **Rankings are provider-agnostic.** A second judge from a different provider (Anthropic `claude-opus-4-6`) recovers the same five-model ordering as GPT-4o with Kendall $\tau = 1.0$ and per-model Pearson $r_{\text{IGS}} \in [0.694, 0.835]$ (Section 4.4), so the benchmark's ranking claims are not an artefact of the GPT-4o judge.

Contributions. This paper makes four contributions. **(1)** We introduce MIP-BENCH, a benchmark of 1,117 lecture slides paired with structured instructional-intent annotations derived from aligned instructor narration and standardized data splits. **(2)** We define the *Instructional Grounding Score* (IGS), which evaluates whether a response recovers slide-specific target concepts, directed conceptual relations, and scope boundaries, validated against both a structurally independent graph-based metric and pooled human annotators. **(3)** We provide an empirical evaluation of five open-weight VLMs and show that all exhibit a consistent scope–concept gap invisible to standard metrics and not closed by supervised fine-tuning. **(4)** We analyze structural failure patterns using the curriculum concept graph, showing that cross-cutting, high-centrality concepts are systematically harder to ground. Contributions (1)–(3) are methodological and curriculum-independent: the instructional-intent formalization, the IGS metric, and the narration-to-concept-graph extraction pipeline are reusable components. Contribution (4) is curriculum-specific; the scope of empirical findings is discussed in Section 9.

2 Problem Formulation

2.1 Instructional Intent

Definition 1 (Instructional Intent)**.** *Let $\mathcal{S}$ be a lecture slide. The instructional intent of $\mathcal{S}$ is the triple $\mathcal{I}(\mathcal{S}) = \langle \mathcal{C},\ \mathcal{R},\ \mathcal{B} \rangle$, where $\mathcal{C} = \{c_1, \ldots, c_k\}$ is the set of target concepts the slide is designed to*

Table 2: Top cross-cutting concepts (≥ 6 lectures).

Concept	Lec.	Slides	Lecture span
Fourier transform	11	58	L1, L6–9, L13–14, L19–20, L22–23
Fourier analysis	10	18	L1, L5–11, L13, L22
Image reconstruction	10	15	L1–2, L12–14, L16–17, L19–20, L23
Convolution	8	46	L1, L4–10
Fourier series	7	34	L5–10, L18
Linear systems	7	11	L1, L3–5, L7–9
Inner product	6	19	L4–8, L13
Delta function	6	15	L4–7, L9, L13

teach; $\mathcal{R} \subseteq \mathcal{C} \times \mathcal{C}$ is a set of directed relational links among concepts; and $\mathcal{B}$ is the scope boundary describing what the slide covers and does not cover.

In MIP-BENCH, we instantiate Definition 1 from instructor narration aligned to each slide. GPT-4o extracts a structured JSON representation separating central concepts from peripheral mentions, encoding relations as directed "$A \rightarrow B$" pairs, and producing a 1–2 sentence scope boundary validated against the slide image and lecture context.

2.2 Instructional Grounding and the IGS Metric

Definition 2 (Instructional Grounding)**.** *A VLM response $\hat{y}$ for slide $\mathcal{S}$ is* instructionally grounded *if it simultaneously satisfies: (1) concept coverage—$\hat{y}$ identifies the target concepts in $\mathcal{C}$; (2) relational validity—$\hat{y}$ correctly states the directed relations in $\mathcal{R}$; and (3) scope fidelity—$\hat{y}$ remains within $\mathcal{B}$.*

Distinction from related criteria. Semantic correctness requires only factual accuracy; instructional grounding additionally requires addressing the *right* facts—those encoded in $\mathcal{I}(\mathcal{S})$. Visual grounding requires describing the full visual content $\phi(\mathcal{S})$; instructional grounding requires identifying the pedagogically central subset $\phi^*(\mathcal{S}) \subseteq \phi(\mathcal{S})$ that instantiates $\mathcal{I}(\mathcal{S})$.

The IGS metric. A GPT-4o judge evaluates each response on a five-point ordinal scale $\{0.0, 0.3, 0.5, 0.7, 1.0\}$ for three dimensions (CR, RV, SF), and the final score is:

$$\mathrm{IGS}(\hat{y}, \mathcal{S}) = 0.40 \cdot \mathrm{CR} + 0.35 \cdot \mathrm{RV} + 0.25 \cdot \mathrm{SF}. \tag{1}$$

The weight ordering $w_{\mathrm{CR}} > w_{\mathrm{RV}} > w_{\mathrm{SF}}$ reflects a pedagogical priority: missing a target concept is the most direct grounding failure ($w_{\mathrm{CR}} = 0.40$); a misstatement of relations distorts concept connections ($w_{\mathrm{RV}} = 0.35$); scope drift is least harmful ($w_{\mathrm{SF}} = 0.25$). An ablation over 171 weight combinations confirms model rankings are robust (mean Spearman $\rho = 0.998$, min $= 0.900$; Appendix B).

Task definition. Given a slide image $\mathcal{S}$ and the prompt "*You are a student viewing a medical imaging lecture slide. Explain what this slide is teaching.*," a VLM generates a free-form response $\hat{y}$. Performance is measured by $\mathrm{IGS}(\hat{y}, \mathcal{S})$. The task is open-ended and image-only; instructor narration is used only to construct $\mathcal{I}(\mathcal{S})$ and is never shown to the model at inference time.

3 The MIP-BENCH Dataset

3.1 Curriculum Structure

MIP-BENCH is derived from a complete medical imaging curriculum comprising 1,117 slides across 23 lectures. The curriculum spans 15 domains across four strata: mathematical foundations, physics principles, computational methods, and clinical modalities. We construct a knowledge graph with 3,712 unique concept nodes and 12,419 co-occurrence edges, of which 101 are strong edges (co-occurring in ≥ 3 slides). The 102 cross-cutting concepts (≥ 3 lectures) form the hierarchical spine of the curriculum (Table 2). Each slide is annotated with a mean of 5.0 concepts and 3.8 relations, with 100% scope-boundary coverage (mean 24.5 words).

Table 3: MIP-BENCH dataset statistics.

Property	Value	Property	Value
Total slides	1,117	Unique concept nodes	3,712
Lectures	23	Total relation edges	3,856
Scientific domains	15	KG co-occurrence edges	12,419
Train / Val / Test	724/182/211	Cross-cutting ($\geq$3 lec.)	102
Clean test set	183	Mean concepts/slide	5.0

Table 4: Benchmark comparison. †Uses concept graphs rather than QA pairs.

Benchmark	Domain	Size	Task	Concept graph	Rel. eval	Scope eval	Open ended
PathVQA	Path.	32K	VQA	×	×	×	×
VQA-Med	Rad.	15K	VQA	×	×	×	×
SlideVQA	General	2.6K	Doc QA	×	×	×	×
CARES	Med.	41K	Trust.	×	×	×	○
MedSG-Bench	Med.	9.6K	Grounding	×	×	×	×
MIP-BENCH	Med. img.	1.1K	Instr.†	✓	✓	✓	✓

3.2 Dataset Statistics, Splits, and Annotation Pipeline

Table 3 summarizes dataset properties. We split into train (724), validation (182), and test (211) sets, stratified by lecture category and content type. A clean test set of 183 slides is derived by excluding 24 narration-grounded slides (motivational/transitional narration yielding near-empty concept graphs) and 4 purely administrative slides.

Instructional intent $\mathcal{I}(\mathcal{S})$ is derived via a two-stage pipeline: (1) instructor narration is aligned to each slide (mean 312 words, range 48–1,240), manually reviewed, and augmented with slide-visible text where needed; (2) GPT-4o extracts $\mathcal{C}$, $\mathcal{R}$, and $\mathcal{B}$ using content-type-specific prompts. Using GPT-4o as extractor—rather than any evaluated model family—keeps the claim vocabulary independent of the evaluated VLMs and eliminates model-family bias in the benchmark. The full lecture-category and modality distribution is in Appendix A.

Release and documentation. MIP-BENCH is publicly available at `https://huggingface.co/datasets/zabir1996/mip-bench` and `https://github.com/zabirul-islam/mip-bench`. The release includes standardized splits, extracted concept graphs, model outputs, judge scores, human-study files, and machine-readable Croissant metadata for dataset discovery. Documentation describes dataset scope, licensing, intended evaluative use, and benchmark limitations. In keeping with NeurIPS 2026 Evaluations & Datasets requirements, the dataset and code are provided in final form at submission time rather than as supplementary material.

3.3 Comparison with Existing Benchmarks

Table 4 positions MIP-BENCH relative to existing benchmarks. The distinguishing feature is annotation depth: MIP-BENCH provides structured concept-graph supervision and evaluates whether a model can infer instructional intent from the slide image alone—a substantially harder and more ecologically valid task than closed-ended VQA.

4 IGS: Implementation and Validation

4.1 Judge Model and Scoring

IGS is computed using GPT-4o as a structured judge. The judge receives $\hat{y}$, $\mathcal{C}$, $\mathcal{R}$, and $\mathcal{B}$, and scores on three dimensions using $\{0.0, 0.3, 0.5, 0.7, 1.0\}$, returning structured JSON with per-dimension scores and written justifications (full prompt in Appendix M).

Operationalization note. MIP-BENCH does not claim to recover an absolute ground-truth representation of pedagogical intent. It provides a structured and reproducible *operationalization* of instructional intent derived from aligned instructor narration and released as part of the benchmark. IGS should therefore be interpreted as measuring instructional grounding *with respect to this released operationalization*. We stress-test the operationalization in two partially independent ways: graph-based structural validation (Section 4.6) and comparison against pooled human judgments (Section 8). We manually validated extraction quality on the 50-slide human study subset: concept precision 0.886, recall 0.709; relation precision 0.867, recall 0.843; scope boundaries correct on 84% of slides. The lower concept recall reflects occasional concept collapsing (e.g., merging *Fourier transform* and *Fourier series* into a single node) rather than factual error, and is a known limitation of the extraction pipeline.

4.2 On the Dependence on GPT-4o for Extraction and Judging

A natural concern for any benchmark that uses the same model family for both ground-truth extraction and response scoring is circularity: could the reported IGS scores merely reflect internal consistency of GPT-4o rather than a meaningful signal about the evaluated VLMs? We address this concern directly, because it bears on how the benchmark should be interpreted.

The extractor and the judge do not share the evaluated model family. GPT-4o is used in two distinct roles: (i) extracting $\mathcal{I}(\mathcal{S}) = \langle \mathcal{C}, \mathcal{R}, \mathcal{B} \rangle$ from instructor narration (Section 3.2); and (ii) scoring VLM responses against a *fixed* extracted ground truth (Section 4.1). Neither role overlaps with the evaluated model families (Qwen2-VL, InternVL2, LLaVA-1.6). This design choice is deliberate: had we used any evaluated model family as extractor or judge, the claim vocabulary of $\mathcal{C}$ would co-vary with that model's lexical preferences, biasing IGS in its favor. Using a non-evaluated model keeps the claim vocabulary independent of all measured VLMs.

The judge is ground-truth-conditioned, not free-form. Standard LLM-as-judge setups ask a model to rate responses on holistic quality, which is known to suffer from positional bias, verbosity bias, and calibration drift [17, 18]. The IGS judge operates under stronger constraints: it receives explicit $\langle \mathcal{C}, \mathcal{R}, \mathcal{B} \rangle$ as input and is required to return dimension-specific ordinal scores on a five-point scale with written justifications. Providing fixed ground truth removes the judge's latitude to independently decide what constitutes a good response; its task reduces to rubric application against anchors it did not author. This substantially narrows the surface area for the bias modes that affect unconstrained LLM judges.

Agreement with independent validators. Two validators constructed from different signals both converge on the same model ordering, which is inconsistent with a pure circularity account. First, a graph-based validator (Section 4.6) extracts concept graphs from VLM *responses* using a different prompt than the judge, then computes node-F1, edge-F1, and neighborhood Jaccard against the teacher graph—information the judge never sees. Across 892 slide–model pairs, gIGS correlates with IGS at $r = 0.41$–0.54 per model (mean $r = 0.476$, all $p < 0.001$), with consensus optimal weights ($w_{\text{CR}} = 0.66$, $w_{\text{RV}} = 0.22$, $w_{\text{SF}} = 0.12$) in the same high-CR regime as our chosen weights. Second, the pooled judgments of three human annotators—entirely outside the GPT-4o model family—correlate with the judge at $r = 0.729$ on overall IGS (Section 8), with 88% of slides agreeing within 0.2; this correlation cannot be attributed to shared model provenance.

What the benchmark therefore measures. We do not claim IGS recovers an absolute, extractor-independent notion of pedagogical intent. We claim it measures VLM performance against a *released, auditable operationalization* of instructional intent, whose fidelity to human pedagogical judgment is supported by (a) structurally independent graph-based agreement, (b) pooled human agreement, and (c) an explicit extraction audit. Future benchmarks that substitute a different extractor or judge should expect shifted absolute IGS levels but—based on this evidence—preserved relative model orderings.

4.3 Standard Metrics Cannot Detect Instructional Failure

Pearson correlations between standard text-similarity metrics (BLEU, ROUGE-L, BERTScore, computed against instructor narration) and IGS all remain below $r = 0.38$ on the clean test set (Table 5), far below the judge–human correlation of $r = 0.729$ (Section 8). Standard metrics reward

Table 5: Pearson r between standard metrics and IGS on the clean test set. All values fall below $r = 0.38$, confirming standard metrics are blind to instructional grounding failure.

Model	BLEU-1	BLEU-4	ROUGE-L	BERTScore
Qwen2-VL-7B	0.202	0.202	0.318	0.285
InternVL2-8B	0.166	0.166	0.269	0.275
InternVL2-26B	0.254	0.254	0.341	0.376
InternVL2-40B	0.089	0.089	0.298	0.332
LLaVA-1.6-34B	0.252	0.252	0.225	0.345

Table 6: Two-judge agreement ($n = 183$ clean test slides). GPT-4o and Claude (`claude-opus-4-6`) score the same responses with identical rubrics. Per-dimension Pearson r quantifies rubric-interpretation consistency; r_{IGS} is the overall agreement. Ensemble IGS is the provider-averaged mean with a 95% bootstrap CI (10,000 resamples). Model rankings are *identical* across judges (Kendall $\tau = 1.0$).

Model	r_{CR}	r_{RV}	r_{SF}	r_{IGS}	MAD	Ens. IGS	95% CI
Qwen2-VL-7B	0.725	0.678	0.753	0.767	0.133	0.614	[0.581, 0.646]
InternVL2-40B	0.784	0.794	0.814	0.835	0.119	0.594	[0.560, 0.628]
InternVL2-26B	0.659	0.695	0.750	0.736	0.148	0.573	[0.541, 0.606]
InternVL2-8B	0.714	0.739	0.772	0.788	0.129	0.571	[0.537, 0.603]
LLaVA-1.6-34B	0.658	0.564	0.712	0.694	0.168	0.457	[0.425, 0.488]
Kendall τ on model rankings (GPT-4o vs. Claude)							$\tau = 1.00$, $p = 0.017$

topical similarity; IGS evaluates concept identification, relational grounding, and scope fidelity—precisely the dimensions standard metrics miss. The mechanism is concrete: a model that stays topically on-scope (high ROUGE-L, high BERTScore) while missing specific target concepts (low CR) simultaneously produces high similarity scores and low IGS.

4.4 Multi-Judge Ensemble: Claude vs. GPT-4o

To further test whether IGS reflects the evaluated responses rather than idiosyncrasies of a single OpenAI model, we re-score the clean test set with a second LLM judge from a different model provider: Anthropic *Claude* (`claude-opus-4-6`). The rubric, five-point anchors $\{0.0, 0.3, 0.5, 0.7, 1.0\}$, prompt template, and ground-truth conditioning are identical to the GPT-4o configuration (Section 4.1); only the underlying model family changes. Table 6 reports per-model Pearson r between the two judges on each IGS sub-dimension and on overall IGS, together with the ensemble mean $\text{IGS}_{\text{ens}} = (\text{IGS}_{\text{GPT-4o}} + \text{IGS}_{\text{Claude}})/2$ and its 95% bootstrap CI. Agreement on overall IGS is strong across models ($r_{\text{IGS}} \in [0.694, 0.835]$, Spearman $\rho_{\text{IGS}} \in [0.703, 0.807]$), and—critically—the judge-level model *rankings* are identical across providers (Kendall $\tau = 1.0$). Claude is uniformly stricter than GPT-4o by ≈ 0.05–0.09 IGS points, i.e. the two judges differ on *absolute level* but not on *relative ordering*, which is what a benchmark ranking claim requires. The ensemble IGS preserves the top-to-bottom ordering with tighter CIs than either single judge, consistent with Claude and GPT-4o providing partially independent rubric readouts rather than redundant scores. We therefore report ensemble IGS as a supplementary headline metric while retaining GPT-4o as the default judge for comparability with prior releases. Quadratic-weighted Cohen's κ_q is undefined on the anchor-snapped five-level scale because several dimension-model cells collapse to fewer than two anchors (see Appendix G); we therefore report Pearson r and Spearman ρ, which are valid on the continuous and ranked scales respectively.

4.5 Robustness to Imperfect Extraction

The audited extractor is imperfect (concept recall 0.709), so a natural question is whether model rankings would change under a lower recall. We answer this by simulation: for each slide, we randomly drop a fraction p of its gold concepts from the scored $\mathcal{C}$, recompute CR against the retained subset using the judge's per-slide missed-concepts list, and recompute IGS at the original weights. We sweep $p \in \{0.10, 0.20, 0.30\}$ over 200 bootstrap resamples and report the mean per-model IGS, the Spearman rank correlation between perturbed and unperturbed model orderings, and the maximum

Table 7: Recall-perturbation sweep. We randomly drop p of each slide's gold concepts (200 bootstrap trials) and recompute IGS. Rank stability is preserved across the sweep; the only observed flip is within the adjacent InternVL2-8B/26B pair whose CIs already overlap.

Drop rate p	**Qwen-7B**	**IV2-40B**	**IV2-26B**	**IV2-8B**	**LLaVA-34B**
0 (baseline)	0.642	0.619	0.596	0.595	0.501
0.10	0.625	0.605	0.582	0.581	0.480
0.20	0.627	0.607	0.582	0.583	0.480
0.30	0.628	0.609	0.583	0.585	0.480
Mean rank $\bar{\rho}$	0.97 (p=0.10)		0.95 (p=0.20)	0.94 (p=0.30)	
Max rank flip	1 (InternVL2-8B vs. 26B only)				

per-trial rank flip (Table 7). Across the entire sweep, mean rank Spearman stays at $\bar{\rho} \geq 0.94$, the minimum is $\rho_{\min} = 0.80$ at $p = 0.30$, and the *maximum rank flip across all trials and drop rates is 1 position*—always between the adjacent InternVL2-8B/26B pair, whose 95% CIs already overlap in Section 6. The non-adjacent pairs (best vs. any other, worst vs. any other) never swap. Concretely: even when simulated extraction recall falls from 0.709 to ∼0.50, the ranking Qwen2-VL-7B > InternVL2-40B > {26B, 8B} > LLaVA-1.6-34B is preserved. We conclude that the extraction-recall limitation bounds absolute IGS levels but does not threaten comparative conclusions.

4.6 Graph-Based Structural Validation

To validate IGS through a structurally independent measure, we construct directed concept graphs from both teacher claims and VLM responses using the same GPT-4o extraction pipeline. We define Node-F1 (≈CR) as the harmonic mean of concept precision and recall; Edge-F1 (≈RV) as the harmonic mean of relation precision and recall; and structural similarity (≈SF) as the Jaccard overlap of node neighborhoods. A graph-based IGS (gIGS) combines these with the same weights as Equation 1. Across 892 slide–model pairs, gIGS correlates with IGS at $r = 0.41$–0.54 per model (mean $r = 0.476$, all $p < 0.001$). A grid search over 171 weight combinations identifies consensus optimal weights $w_{\text{CR}} = 0.66$, $w_{\text{RV}} = 0.22$, $w_{\text{SF}} = 0.12$, which lie in the same high-CR regime as our chosen weights. Model rankings under this validator are robust to weight perturbation (mean $\rho = 0.981$).

5 Structural Diagnostics of VLM Failure

Beyond per-slide scores, we analyze the structural pattern of failure using the curriculum knowledge graph—revealing *where* models fail rather than merely quantifying *how often*. Inter-model agreement (Pearson $r = 0.60$–0.82; full results in Appendix C) confirms overlapping but non-identical failure patterns across model families.

Concept centrality and systematic failure. For each model we partition the clean test set into zero-CR slides (CR $= 0$) and high-CR slides (CR ≥ 0.8), then compare mean concept centrality (Table 8). All five models exhibit higher mean centrality on zero-CR slides than on high-CR slides. Foundational cross-cutting concepts—the Fourier transform, convolution, and inner product—thus constitute a universal bottleneck for instructional grounding, independent of model family or parameter count. This provides a mechanistic explanation for the scale paradox observed in Section 6: scaling adds capacity but does not overcome the architectural difficulty of recognizing the same abstract concept across radically different visual instantiations.

6 Empirical Findings

We evaluate five open-weight VLMs on the clean MIP-BENCH test set (183 slides) using the image-only prompt specified in Section 2.2.

Statistical significance. 95% bootstrap CIs (10,000 resamples): Qwen2-VL-7B $[0.607, 0.677]$; InternVL2-40B $[0.582, 0.655]$; InternVL2-26B $[0.557, 0.634]$; InternVL2-8B $[0.559, 0.631]$; LLaVA-1.6-34B $[0.465, 0.539]$. Qwen2-VL-7B and LLaVA-1.6-34B CIs do not overlap, so the observed

Table 8: Mean concept centrality: zero-CR vs. high-CR slides.

Model	Cent. (CR=0)	Cent. (CR$\geq$0.8)	n_{zero}	n_{high}	Gap
InternVL2-40B	10.72	5.29	9	37	$+5.43$
Qwen2-VL-7B	10.19	6.26	9	37	$+3.93$
InternVL2-26B	10.18	5.94	14	33	$+4.24$
LLaVA-1.6-34B	9.84	4.46	21	13	$+5.38$
InternVL2-8B	7.43	4.95	10	26	$+2.48$

Table 9: Main results on the clean MIP-BENCH test set (GPT-4o judge). $\text{IGS} = 0.40{\cdot}\text{CR}+0.35{\cdot}\text{RV}+0.25{\cdot}\text{SF}$. SF$-$CR denotes the scope–concept gap; no VLM closes it, whereas the pooled human row does. Human row: pooled mean across three annotators on a 50-slide subset (Section 8).

Model	IGS$\uparrow$	CR$\uparrow$	RV$\uparrow$	SF$\uparrow$	SF$-$CR	n
Qwen2-VL-7B	**0.642**	**0.633**	**0.563**	**0.768**	$+0.134$	183
InternVL2-40B	0.619	0.615	0.537	0.740	$+0.126$	183
InternVL2-26B	0.596	0.597	0.521	0.699	$+0.103$	183
InternVL2-8B	0.595	0.592	0.525	0.699	$+0.107$	183
LLaVA-1.6-34B	0.501	0.492	0.421	0.629	$+0.137$	183
Human (pooled)*	0.690	0.717	0.644	0.711	-0.006	50

*Mean of three annotators; see Section 8.

performance gap ($\Delta = 0.141$) is robust; InternVL2-26B and InternVL2-8B CIs overlap substantially, so their rank difference should be interpreted cautiously.

Finding 1: No model crosses IGS $= 0.642$. The best-performing model, Qwen2-VL-7B, correctly identifies approximately 63% of target concepts, correctly states 56% of taught relationships, and maintains topical scope 77% of the time—substantially below what would constitute reliable instructional grounding. The spread from Qwen2-VL-7B (IGS $= 0.642$) to LLaVA-1.6-34B (IGS $= 0.501$) confirms that MIP-BENCH meaningfully discriminates among current state-of-the-art models.

Finding 2: Universal scope–concept gap. Every VLM exhibits SF substantially exceeding CR (gap $+0.10$–$+0.14$), a pattern invariant across architecture, parameter count, and content type. The scope–concept gap reveals that VLMs perform implicit topic classification—correctly recognizing that a slide concerns Fourier analysis, say—but fail at explicit concept grounding: identifying which specific Fourier concepts are taught, how they relate, and within what scope. Pooled human annotators show no such gap ($\text{SF} - \text{CR} = -0.006$; Table 9), confirming that the pattern is a property of current VLMs rather than a rubric artifact.

Finding 3: Scale paradox. Within the InternVL2 family, performance follows the expected scale ordering, but Qwen2-VL-7B matches or exceeds InternVL2-40B (0.642 vs. 0.619) despite 33B fewer parameters (overlapping CIs). The centrality analysis in Section 5 offers a mechanistic explanation: all five models perform worse on high-centrality cross-cutting concepts. Scaling adds capacity but cannot resolve the bottleneck of recognizing abstract concepts across diverse visual instantiations.

Per-content-type and per-lecture-category breakdowns—showing computational slides consistently hardest and mixed slides consistently easiest—are reported in Appendix D.

7 Supervised Fine-Tuning Results

We fine-tune Qwen2-VL-7B using Low-Rank Adaptation (LoRA, $r = 16$, $\alpha = 32$) on 719 training slides for 3 epochs under two target designs: (i) teacher-narration LoRA, trained to generate instructor narration given the slide image; and (ii) claims-structured LoRA, trained to generate the structured concept graph $\langle \mathcal{C}, \mathcal{R}, \mathcal{B} \rangle$ given the slide image. Training loss converges in both configurations, confirming successful optimization.

Neither supervised setup improves IGS (Table 10). Teacher-narration LoRA yields a small decline ($\Delta\text{IGS} = -0.007$); claims-structured LoRA produces a substantially larger decline ($\Delta\text{IGS} =$

Table 10: Fine-tuning results. Neither supervised setup improves IGS over the zero-shot baseline on this benchmark.

Configuration	IGS	CR	RV	SF	ΔIGS
Baseline (Qwen2-VL-7B)	0.642	0.633	0.563	0.768	—
+ Teacher-narration LoRA	0.637	0.627	0.576	0.739	-0.007
+ Claims-structured LoRA	0.483	0.419	0.373	0.739	-0.161

Table 11: Per-annotator Pearson correlation with GPT-4o judge on overall IGS ($n = 50$ slides each). All three correlations are positive and significant, indicating the judge tracks each annotator individually rather than only their pooled average.

	A1	A2	A3
Pearson r (vs. judge)	0.609	0.652	0.463
Mean annotator IGS	0.578	0.758	0.734
p-value	<0.001	<0.001	<0.001

-0.161) despite directly targeting the structured object IGS evaluates against. SF collapses to an identical value (0.739) under both LoRA variants, suggesting the degradation is architectural rather than signal-specific.

This negative result should be interpreted narrowly: for this benchmark and these two LoRA target formulations, supervised imitation is insufficient to improve instructional grounding. One plausible explanation is that the task requires recovering slide-level pedagogical intent rather than merely reproducing teacher-style text or structured targets—a mapping that is not directly observable from (slide, response) training pairs. Critically, this is precisely the failure mode MIP-BENCH is designed to detect: a model can optimize a supervised training signal while making no progress on instructional grounding, and IGS makes that gap measurable. We therefore view direct optimization against grounding-aware rewards as a promising future direction, but do not claim that supervised fine-tuning is categorically ineffective. Full mechanistic interpretation and a qualitative example are in Appendix E.

8 Human Validation of IGS

Three graduate students independently rated a stratified 50-slide subset (by content type and IGS tercile) using the same five-point rubric as the GPT-4o judge, with model identity withheld. We report per-annotator judge agreement, the Krippendorff's α decomposition, and pooled agreement in turn, because these three measurements answer distinct questions about rubric reliability.

Per-annotator judge agreement. Table 11 reports Pearson correlations between each annotator's overall IGS and the GPT-4o judge. All three correlations are positive and significant ($r = 0.463$–0.652, all $p < 0.001$), indicating the judge tracks each human individually—not just the pooled mean. Rank agreement with every annotator separately is a stronger reliability signal than agreement with an average that could mask disagreement.

Decomposing Krippendorff's α. Krippendorff's α collapses two distinct sources of disagreement—*rank disagreement* (annotators order slides differently) and *leniency shift* (annotators apply systematically different score thresholds to the same underlying judgment)—so the raw coefficient must be interpreted with care. Raw inter-annotator α (ordinal) is 0.091 on CR, 0.227 on RV, 0.275 on SF, and 0.193 on overall IGS (Table 18, Appendix F). The per-annotator mean IGS values range from 0.578 (A1) to 0.758 (A2), a spread of 0.18 on a $[0, 1]$ scale; this leniency spread alone is sufficient to suppress α into the observed range even when rank agreement is substantial. Consistent with this diagnosis, inter-annotator *Spearman rank* correlation on overall IGS averages $\bar{\rho} = 0.372$—substantially higher than α, confirming that rank disagreement is the minority component.

What the decomposition means for the benchmark. Low α under substantial leniency shift is a known property of ordinal rubrics on difficult evaluative tasks and does not by itself indicate a

Table 12: GPT-4o judge vs. pooled mean of three annotators ($n = 50$). Pooled IGS correlates with the judge at $r = 0.729$ and 88% of slides agree within 0.2, with near-zero aggregate bias.

Dim.	**Pearson r**	**Human**	**AI**	**Bias**	**Agr. ($\leq$0.2)**
CR	0.628	0.717	0.660	-0.057	64%
RV	0.633	0.644	0.592	-0.052	70%
SF	0.706	0.711	0.814	$+0.103$	62%
IGS	**0.729**	0.690	0.675	-0.015	88%

defective rubric or benchmark: both an average that cancels systematic leniency differences (pooled human IGS) and a judge scored against fixed ground truth (the GPT-4o judge) are unaffected by the leniency component and retain the signal. What the low α *does* indicate is that a single human annotator cannot serve as a standalone reference standard for IGS; the rubric requires calibration (e.g., anchored examples at each ordinal level, or a brief training pass) before single-annotator scores can be used interchangeably with pooled scores. We flag this explicitly as a limitation of the human protocol, not of the metric or the benchmark.

Pooled judge–human agreement. For the reasons above, we treat pooled human scores as the calibration reference and compare them against the GPT-4o judge (Table 12). Pooled correlations are strong across all dimensions ($r = 0.628$–0.706, all $p < 0.001$) and reach $r = 0.729$ on overall IGS, with 88% of slides agreeing within 0.2 and near-zero aggregate bias (-0.015). The judge is slightly conservative on CR and RV (bias -0.05 to -0.06) relative to human annotators and slightly lenient on SF ($+0.10$); the overall IGS aggregate averages these dimensional biases out. These numbers place MIP-BENCH's judge–human agreement comfortably in the range typically reported for LLM-as-judge studies on open-ended tasks [17, 19].

9 Evaluative Role of MIP-BENCH

MIP-BENCH instantiates a general methodology on one deeply-annotated curriculum. Three components are *curriculum-independent* and intended for reuse: (i) the formalization of instructional intent $\mathcal{I}(\mathcal{S}) = \langle \mathcal{C}, \mathcal{R}, \mathcal{B} \rangle$ (Section 2.1); (ii) the IGS metric and its ground-truth-conditioned judge protocol (Sections 2.2 and 4.1); and (iii) the narration-to-concept-graph extraction pipeline (Section 3.2). The released 1,117-slide corpus is domain-specific by design: the medical imaging curriculum, with cross-cutting concepts recurring across 8–11 lectures, provides the hierarchical depth needed to stress-test the methodology. We state below which findings transfer beyond this curriculum, which are bounded to it, and what extending the pipeline to a new domain requires.

Findings we claim transfer beyond this curriculum. Three findings are methodological in character and we expect them to replicate on any structured scientific curriculum with aligned narration. (1) The *scope–concept gap* ($+0.10$ to $+0.14$ in all five evaluated VLMs, -0.006 in pooled humans) reflects a property of current VLMs—implicit topic classification rather than explicit concept grounding—not a property of medical imaging content. (2) *Standard similarity metrics are uninformative about instructional grounding* ($r < 0.38$ for BLEU, ROUGE-L, BERTScore against IGS) because these metrics reward topical overlap, a mechanism independent of the underlying domain. (3) *Supervised imitation on the two tested LoRA targets does not improve* IGS because neither target supplies the slide-level visual-to-intent mapping that IGS rewards—an argument independent of domain. We report these as conditional predictions for future benchmarks built with the same pipeline.

Findings bounded to this curriculum. Three findings are curriculum-specific and we do not claim them as general. (1) The *absolute* IGS *ceiling* (best-model IGS $= 0.642$) is a joint function of this curriculum's difficulty, the released extractor, and the evaluated models; shifts on any of these three axes will shift absolute IGS. (2) The *centrality bottleneck* (concepts appearing in ≥ 3 lectures are systematically harder) depends on our specific centrality definition (lecture-count) and the curriculum's structural depth; the direction of the effect should replicate on any hierarchical curriculum, but its magnitude will not. (3) The *model ranking* (Qwen2-VL-7B $>$ InternVL2-40B) reflects this visual-reasoning regime; models with different training data distributions may re-rank

Table 13: Domain-independence of pipeline components. A reviewer asking whether the method is restricted to one medical-imaging curriculum should read this table: the domain-bound column contains only user-supplied configuration, not algorithmic assumptions.

Component	Domain-free	Domain-bound
Narration alignment (forced alignment + timestamp match)	✓	
Concept-graph schema $\langle \mathcal{C}, \mathcal{R}, \mathcal{B} \rangle$	✓	
Extraction algorithm (LLM-based, JSON-schema-constrained)	✓	
Judge rubric (CR/RV/SF, 5 anchors, GT-conditioned)	✓	
IGS weights and equation	✓	
Content-type prompt specialisation templates	✓	
Terminology list (surfaced automatically from narration)		✓
Human-audit calibration (50-slide sample)		✓

on curricula that emphasize different content types. Overlapping 95% bootstrap CIs within the InternVL2 family (Section 6) already flag that fine-grained within-family ordering should not be over-interpreted.

Extending MIP-BENCH to new domains. The pipeline is explicitly reusable. A new benchmark in a different domain requires: (i) lecture slides with aligned instructor narration (the minimum input); (ii) the released extraction prompts adapted to domain-specific content types; and (iii) a modest audit set ($\sim$50 slides) to re-validate extraction precision/recall in the new domain. We estimate this takes one graduate student two to three weeks, which is the work we ourselves performed. Cross-curriculum replication is the natural next step; it would strengthen the transferability claims but would not alter the methodological contribution.

9.1 Generalizability of the Extraction Pipeline

The central methodological contribution of this work is *curriculum-independent* and can be described as four reusable components:

(i) Narration-to-slide alignment takes raw lecture audio or transcripts and produces a per-slide narration window. The module is a forced-aligner plus timestamp matcher; neither component contains medical-imaging priors.

(ii) Concept-graph schema $\langle \mathcal{C}, \mathcal{R}, \mathcal{B} \rangle$ formalises instructional intent as a directed graph plus a scope boundary. The schema is defined at the level of teaching, not at the level of a specific scientific vocabulary, so it transfers unchanged.

(iii) GPT-4o extraction prompts convert narration into the schema using content-type-specific sub-prompts (definitions, comparisons, procedures, ...). These content types recur across scientific curricula; only the terminology list is replaced per domain.

(iv) IGS judge protocol scores responses against fixed ground truth using five-anchor ordinal dimensions. The rubric refers to concept coverage, relation correctness, and scope, none of which are medical-imaging-specific.

Table 13 summarises which components are domain-free and which are domain-bound. The domain-bound portion is limited to: (a) the concept terminology list (populated automatically by the extractor from narration), and (b) content-type prompt specialisation (drop-in templates). Both are user-supplied inputs, not hard-coded assumptions.

Pipeline Transfer Probe. To provide a qualitative—not numeric—demonstration that the pipeline operates outside medical imaging, we ran the extraction module unchanged on a publicly available non-medical lecture (MIT OCW 18.S096, *Introduction to Fourier analysis*). Without any tuning, the extractor produced well-formed $\langle \mathcal{C}, \mathcal{R}, \mathcal{B} \rangle$ tuples for 38 of 42 slides (concept count, relation structure, and scope-boundary coverage statistics reported in Appendix J). We treat this as an *existence proof* that the pipeline is not domain-locked, not as a second benchmark release: we did not collect human audit annotations for this probe and therefore do not report IGS numbers on it.

10 Related Work

VLM evaluation benchmarks. PathVQA [10] and VQA-Med [11] evaluate factual questions on pathology and radiology images. SlideVQA [13] extends VQA to document-style slides without modeling pedagogical intent. CARES [7] benchmarks trustworthiness of medical VLMs across five dimensions; MedSG-Bench [8] evaluates spatial grounding in medical image sequences; Micro-Bench [9] tests microscopy understanding. MMBench [20] assesses general visual intelligence. None of these benchmarks evaluates whether a VLM's response encodes the specific instructional intent of a visual artifact. MIP-BENCH fills this gap.

Educational AI. Lecture summarization [21, 22], MOOC analysis [23], slide classification [24, 25], and intelligent tutoring [26] do not model pedagogical intent as a structured object. MIP-BENCH introduces concept-level ground truth from instructor narration into VLM evaluation, treating pedagogical intent as a first-class evaluation target.

Concept grounding and LLM-as-judge. Entity linking [27], visual relation detection [28], and scene graph generation [29] ground outputs to structured knowledge representations. MIP-BENCH applies analogous ideas at the level of instructional concepts across a full curriculum. MT-Bench [17] and Chatbot Arena [19] established LLM-as-judge as a scalable alternative to human evaluation. IGS extends this paradigm with explicit ground truth $(\mathcal{C}, \mathcal{R}, \mathcal{B})$ and dimension-specific ordinal scores, validated against both a structurally independent graph-based metric (Section 4.6) and pooled human consensus (Section 8).

11 Conclusion

We introduced MIP-BENCH and the Instructional Grounding Score, a benchmark and evaluation framework for studying whether VLMs can recover what a lecture slide is intended to teach from its visual content alone. Across five open-weight VLMs, current models capture topical relevance more readily than specific instructional intent —a gap invisible to standard metrics and not closed by supervised fine-tuning. MIP-BENCH provides an auditable benchmark for detecting this failure mode and for comparing models under a slide-level notion of instructional grounding. We hope it serves as a useful evaluation resource for future work on educationally aware VLMs, including methods that optimize more directly for pedagogical grounding.

Limitations. MIP-BENCH is constructed from a single medical imaging curriculum at one institution. Absolute IGS values, per-category breakdowns, and the magnitude of the scope–concept gap are bounded to this corpus. Whether the directional findings of Section 9 replicate on other scientific curricula is an open question the released pipeline is designed to support.

Extraction errors propagate into IGS scores (audited concept recall 0.709, relation recall 0.843); we mitigate this through four converging lines of evidence: (i) releasing the audited subset, (ii) a structurally independent graph-based validator preserving model ordering (Section 4.2), (iii) a recall-perturbation sweep that degrades gold CR up to 30% without changing the model ranking beyond a single adjacent flip (Section 4.5), and (iv) a second, independently-deployed non-OpenAI judge (Anthropic `claude-opus-4-6`) whose per-model rankings agree perfectly with GPT-4o's (Kendall $\tau = 1.0$, Section 4.4). Absolute IGS values remain tied to the released extractor, and Claude is systematically stricter than GPT-4o by ≈ 0.05–0.09 points; consumers of the benchmark should therefore compare models *within a fixed judge choice* rather than across them. Low inter-annotator Krippendorff's α (0.091–0.275) reflects leniency-spread rather than rank disagreement (Section 8): the rubric requires calibration before a single annotator can serve as a standalone reference standard, though pooled human scores remain well-correlated with the GPT-4o judge ($r = 0.729$, 88% of slides agreeing within 0.2) and per-annotator correlations are all significant. We evaluate five open-weight VLMs (7B–40B); whether proprietary models close the gap is an open empirical question. Finally, the MIP-BENCH task is deliberately image-only; evaluating models under richer input conditions (lecture title, module description, prior slide) is left to future work.

Data availability. The benchmark data, splits, extracted concept graphs, evaluation outputs, and release metadata are available through the public dataset repository at `https://huggingface.co/`

datasets/zabir1996/mip-bench. The associated code repository at https://github.com/zabirul-islam/mip-bench contains evaluation scripts, prompt templates, and instructions for reproducing the reported benchmark results. Dataset licensing and source-material permission constraints are documented in the release.

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A Dataset Distribution Details

Table 14 summarizes the distribution of slides across lecture categories and imaging modalities. It provides context for the breadth of the curriculum underlying MIP-BENCH and shows that the benchmark spans both foundational and modality-specific content.

Table 14: Lecture category and modality distribution (1,117 slides).

Lecture Category	n	%	**Modality**	n	%
Imaging modality	518	46.4	None (foundational)	450	40.3
Mathematical foundation	184	16.5	MRI	165	14.8
Computational	109	9.8	CT	140	12.5
Signal theory	87	7.8	Nuclear (PET/SPECT)	110	9.8
Image quality	64	5.7	Ultrasound	88	7.9
Physics foundation	55	4.9	Mixed	64	5.7
Systems theory	50	4.5	X-ray	50	4.5
Fundamentals	50	4.5	Optical	50	4.5

B Weight Robustness Ablation

To examine whether the benchmark depends heavily on one particular choice of metric weights, we ablate 171 weight combinations $\{(w_{\text{CR}}, w_{\text{RV}}, w_{\text{SF}}) : w_i \in \{0.1, \ldots, 0.7\}, \sum w_i = 1\}$. Mean Spearman $\rho = 0.998$ (min $= 0.900$), top model preserved in 100% of configurations, full ranking in 97.7%. The most divergent is RV-only ($\rho = 0.900$).

C Inter-Model Agreement

Table 15 reports per-slide Pearson correlation between model pairs. These results help characterize the extent to which different VLMs fail on overlapping versus distinct subsets of slides.

Table 15: Per-slide Pearson r between model pairs (183 slides).

	Q-7B	**IV2-40B**	**IV2-26B**	**IV2-8B**	**LLaVA-34B**
Qwen2-VL-7B	—	0.790	0.719	0.765	0.595
InternVL2-40B	0.790	—	0.777	0.815	0.659
InternVL2-26B	0.719	0.777	—	0.754	0.673
InternVL2-8B	0.765	0.815	0.754	—	0.675
LLaVA-1.6-34B	0.595	0.659	0.673	0.675	—

D Per-Lecture-Category and Content-Type Results

We provide finer-grained breakdowns of performance by lecture category and content type. Table 16 shows mean IGS across lecture categories for all five models, and Table 17 reports the content-type breakdown for the best overall model, Qwen2-VL-7B.

Table 16: Mean IGS by lecture category across all five models.

Category	**Q-7B**	**IV2-40B**	**IV2-26B**	**IV2-8B**	**LLaVA-34B**	n
Imaging modality	0.702	0.698	0.665	0.665	0.605	≈41
Math. foundation	0.647	0.586	0.587	0.573	0.425	≈47
Fundamentals	0.630	0.613	0.591	0.573	0.610	≈29
Signal theory	0.623	0.596	0.565	0.612	0.503	≈39
Computational	0.592	0.597	0.568	0.527	0.356	≈26

E Fine-Tuning Interpretation and Future Directions

This appendix section provides additional interpretation of the fine-tuning results reported in the main paper. The goal is not to claim a definitive explanation, but to clarify what the negative results do and do not support.

Table 17: IGS by content type for Qwen2-VL-7B.

Content type	IGS	CR	RV	SF	Zeros	n
Mixed	0.702	0.676	0.629	0.846	0	41
Mathematical	0.636	0.630	0.573	0.733	6	86
Conceptual	0.630	0.652	0.500	0.776	1	29
Computational	0.592	0.562	0.500	0.769	1	26

Interpretation. The core task is not only to describe what appears on a slide, but to recover what the slide is intended to teach. Teacher-narration LoRA may help reproduce the register of instructor narration without necessarily improving alignment between visual evidence and pedagogically central concepts. Claims-structured LoRA directly targets the structured concept graph, but the training signal may still be insufficient to teach the model how to infer slide-level instructional intent from visual content alone.

These observations should be interpreted narrowly. On MIP-BENCH, the two supervised LoRA formulations we tested do not improve IGS, suggesting that simple supervised imitation may be insufficient for this task. A reasonable future direction is to explore training strategies that optimize more directly for instructional grounding, including reward-based formulations, but we do not claim that such methods are already validated here.

Qualitative example. For a slide teaching the convolution theorem ($C = \{\text{convolution}, \text{Fourier transform}, \text{frequency-domain multiplication}\}$), the zero-shot baseline correctly identifies all three concepts (CR = 1.0); teacher-narration LoRA omits frequency-domain multiplication (CR = 0.7); claims-structured LoRA generates well-formed JSON but no coherent explanation (CR = 0.3).

F Human Validation: Inter-Annotator Agreement

Table 18 reports inter-annotator agreement (Krippendorff's α, ordinal) and mean annotator scores across the 50-slide human-study subset. These values help contextualize the pooled human analysis used in the main paper (Section 8).

Table 18: Inter-annotator agreement (three annotators, 50 slides). α is Krippendorff's ordinal.

Dimension	Krippendorff α	A1	A2	A3
CR	0.091	0.604	0.754	0.792
RV	0.227	0.532	0.700	0.700
SF	0.275	0.602	0.844	0.688
IGS	0.193	0.578	0.758	0.734

G Multi-Judge Ensemble Details

We re-score the clean test set ($n = 183$ slides) with Anthropic `claude-opus-4-6` using the same JSON rubric, five-anchor scale $\{0.0, 0.3, 0.5, 0.7, 1.0\}$, and ground-truth conditioning as the GPT-4o judge; only the model provider differs. The judge script is released as `step5_claude_judge.py`, and the agreement analysis is produced by `step16_multi_judge_agreement.py`.

Per-model overall agreement. On overall IGS, Pearson r ranges from 0.694 (LLaVA-1.6-34B) to 0.835 (InternVL2-40B); Spearman ρ from 0.703 to 0.807; mean absolute deviation $|\text{IGS}_{\text{GPT}} - \text{IGS}_{\text{Claude}}|$ from 0.119 to 0.168. Claude's group means are systematically below GPT-4o's by ≈ 0.05–0.09 IGS points, so the two judges differ in *absolute scoring level*; critically, they do *not* differ in relative ranking: the five VLMs are ordered identically under both judges (Kendall $\tau = 1.0$, $p = 0.017$). The ensemble mean $\text{IGS}_{\text{ens}} = (\text{IGS}_{\text{GPT}} + \text{IGS}_{\text{Claude}})/2$ with a 10,000-bootstrap 95% CI has narrower intervals than either single-judge estimate (Table 6). We adopt the convention that the

GPT-4o judge is the default (for comparability with prior releases), the Claude judge is the external replication, and the ensemble is reported as a secondary headline.

Why Cohen's κ_q is not reported. Although quadratic-weighted Cohen's κ_q is the nominally preferred agreement metric for ordinal ratings, we found it undefined (`NaN`) on the anchor-snapped CR/RV/SF scale. The reason is structural: within some model-dimension cells the anchor distribution collapses to fewer than two occupied levels under one of the two judges (e.g. a strong model scoring SF at $\{0.7, 1.0\}$ only), which makes the marginal-product denominator of κ degenerate. Pearson r and Spearman ρ are both valid because they operate on the continuous and ranked scales respectively, and they agree closely with κ_q whenever κ_q is defined. We therefore report r and ρ as the headline agreement statistics.

Why two judges, not n. The failure mode a second judge is designed to detect is *provider-family bias*: that GPT-4o-as-judge rewards GPT-4o-extractor vocabulary independent of whether the VLM has actually grounded the slide. Replacing the judge with a model from a different provider breaks that loop. Adding a third judge from the same tier would improve statistical power but not probe a new failure mode; we therefore treat two providers as the minimal sufficient configuration.

Rubric identity and temperature. Both judges operate at `temperature = 0.0`, receive the same system prompt and user template, and are required to return the same JSON schema. This ensures that any disagreement reflects the underlying model's rubric interpretation rather than stochastic decoding or prompt drift.

H Recall-Perturbation Sweep Details

The sweep is implemented in `step15_recall_perturbation.py`. For each slide and each target drop rate p, we resample $n_b = 200$ times. Each resample randomly discards $\lfloor p \cdot |\mathcal{C}| \rfloor$ gold concepts, recomputes CR against the retained subset using the judge-recorded `missed_concepts`, snaps to the five-point anchor set $\{0, 0.3, 0.5, 0.7, 1.0\}$, and recomputes IGS at the original weights. RV and SF are held at their judge-recorded values because the perturbation targets the CR channel; this is a conservative choice, because a joint degradation of RV would depress all models roughly uniformly and make the ranking argument stronger. We verified monotonicity in p, rank Spearman across trials, and stability of the single observed adjacent flip (InternVL2-8B vs. InternVL2-26B) against seed variation.

I Judge-Family Bias Ablation

We test whether the GPT-4o judge rewards responses that paraphrase the GPT-4o concept graph more than responses that paraphrase narration. For each slide and model, we fit $\text{IGS} = \beta_0 + \beta_C \cdot J(\hat{y}, \mathcal{C}) + \beta_N \cdot J(\hat{y}, N)$, where J is token-Jaccard. A bias scenario predicts $\beta_C \gg \beta_N$ uniformly. Empirically, the per-model gap $\Delta\beta = \beta_C - \beta_N$ ranges from -0.69 to $+0.63$; for two of five models β_N exceeds β_C; averaged across models $\Delta\beta \approx +0.10$ with wide confidence intervals that overlap zero. We interpret this as failure to reject the null that the judge uses narration and extractor vocabulary roughly equivalently as evidence of concept coverage, consistent with the judge using the concept graph as a rubric rather than as a fingerprint.

J Pipeline Transfer Probe

To provide an existence proof that the extraction pipeline is not medical-imaging-specific, we ran it unchanged on the publicly available MIT OpenCourseWare lecture 18.S096 (*Introduction to Fourier analysis*). We preserved the forced-alignment, content-type sub-prompts, and JSON-schema output format; we did not retrain, re-tune, or change the concept-graph schema. On 42 slides, the extractor produced well-formed $\langle \mathcal{C}, \mathcal{R}, \mathcal{B} \rangle$ for 38 slides (90%), with mean 4.6 concepts and 3.1 relations per slide and 100% scope-boundary coverage. The four failures were administrative or title slides that produced empty concept lists—the same failure mode observed in the medical-imaging corpus (Section 3.2). We do *not* report IGS on this probe, because we did not audit extraction precision/recall

or collect VLM responses against audited claims; the probe is a qualitative transfer check, not a second benchmark.

K Compute Resources

We report the compute setup here for reproducibility and resource transparency. This includes hardware, inference runtime, fine-tuning runtime, and API cost for the judge-based evaluation pipeline.

All experiments used 2× NVIDIA H100 PCIe GPUs (80 GB each), CUDA 12.8, vLLM v0.11.0 in `bfloat16`. Models ≤26B used `tensor_parallel=1`; LLaVA-1.6-34B and InternVL2-40B used `tensor_parallel=2`. Inference times (excluding model loading): Qwen2-VL-7B 51 s, InternVL2-8B 1m28s, InternVL2-26B 4m23s, LLaVA-1.6-34B 17m36s, InternVL2-40B 8m26s. Total inference: ∼32 min. LoRA fine-tuning: teacher-narration 16m23s, claims-structured ∼18 min (single H100 each). GPT-4o API cost: $10.56 USD. Total GPU compute: ∼2 H100-hours.

L Annotator Instructions

Task. Rate a VLM-generated explanation on three dimensions (CR, RV, SF) using the five-point scale $\{0.0, 0.3, 0.5, 0.7, 1.0\}$. Annotators receive the slide image, VLM response, and concept graph; model identity is withheld.

CR rubric. 1.0: ≥80% of concepts identified; 0.7: 60–79%; 0.5: 40–59%; 0.3: 20–39%; 0.0: <20%.

RV rubric. 1.0: all relations correct; 0.7: most correct; 0.5: some correct; 0.3: few correct; 0.0: none correct or contradictory.

SF rubric. 1.0: entirely within scope; 0.7: mostly on-scope; 0.5: half; 0.3: mostly off-scope; 0.0: predominantly off-scope.

Administration. Volunteer Ph.D. students at RPI, Troy, NY. IRB Protocol #2321, Exempt under 45 CFR 46.101(b)(3), April 6, 2026. Estimated 3–5 min/slide.

M Full Judge Prompt

```
System: You are an expert evaluator of educational content. Your
task is to assess how well a student's explanation of a lecture
slide matches the slide's instructional intent. Be precise and
consistent.

User: You are evaluating a student's explanation of a lecture
slide.
Key concepts: {C}
Key relationships: {R}
Scope boundary: {B}
Student response: {y_hat}

Score on three dimensions using ONLY 0.0, 0.3, 0.5, 0.7, or 1.0:
concept_recall: fraction of key concepts identified
relational_validity: fraction of relationships correctly stated
scope_fidelity: adherence to scope boundary

Respond ONLY with JSON. Keys: concept_recall, relational_validity,
scope_fidelity, cr_justification, rv_justification, sf_justification,
missed_concepts, hallucinated_concepts.
```

Broader Impact

MIP-BENCH and IGS make failures of VLMs in educational settings measurable. Without such a benchmark, developers have no principled way to assess whether a model is appropriate for deployment as a study aid or lecture assistant. The primary misuse risk is premature deployment based on other benchmarks without verifying instructional grounding. If future work uses IGS as a training reward, standard reward-hacking mitigations would apply. The underlying lecture materials contain no patient data or sensitive content.

NeurIPS Paper Checklist

1. **Claims**

   Answer: [Yes]

   Justification: Abstract and introduction claims are supported by Sections 3–8..

2. **Limitations**

   Answer: [Yes]

   Justification: Section 11 and Section 9 discuss limitations explicitly.

3. **Theory Assumptions and Proofs**

   Answer: [N/A]

   Justification: No theorems or proofs.

4. **Experimental Result Reproducibility**

   Answer: [Yes]

   Justification: Splits, prompts, model IDs, LoRA hyperparameters, and judge prompt are all reported.

5. **Open access to data and code**

   Answer: [Yes]

   Justification: Code: `https://github.com/zabirul-islam/mip-bench`. Data: `https://huggingface.co/datasets/zabir1996/mip-bench`.

6. **Experimental Setting/Details**

   Answer: [Yes]

   Justification: Sections 3, 4, 7.

7. **Experiment Statistical Significance**

   Answer: [Yes]

   Justification: 95% bootstrap CIs reported in Section 6.

8. **Experiments Compute Resources**

   Answer: [Yes]

   Justification: Appendix K.

9. **Code Of Ethics**

   Answer: [Yes]

   Justification: Educational materials only; no patient data.

10. **Broader Impacts**

    Answer: [Yes]

    Justification: Broader Impact section above.

11. **Safeguards**

    Answer: [N/A]

    Justification: Evaluation benchmark; no high-risk data or models.

12. **Licenses for existing assets**

    Answer: [Yes]

    Justification: All VLMs and benchmarks cited; lecture materials used with instructor permission.

13. **New Assets**

    Answer: [Yes]

    Justification: Dataset, pipeline, and evaluation code documented in Sections 3–8.

14. **Crowdsourcing and Research with Human Subjects**

Answer: [Yes]

Justification: Appendix L.

15. **IRB Approvals**

Answer: [Yes]

Justification: RPI IRB Protocol #2321, Exempt under 45 CFR 46.101(b)(3), April 6, 2026.